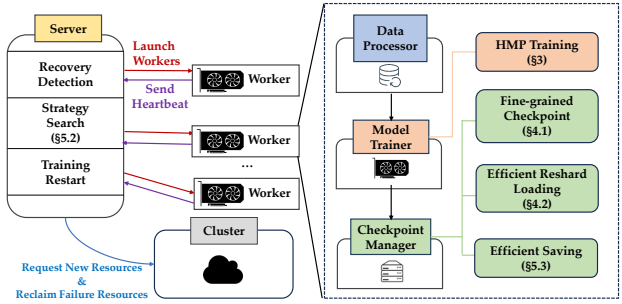
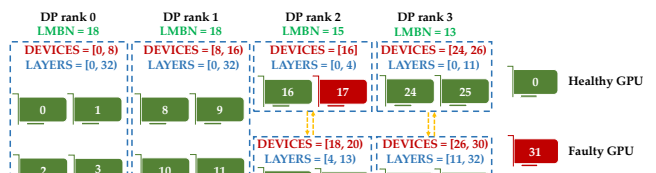

Server
Recovery Detection
Strategy Search (§5.2)
Training Restart
Launch Workers
Send Heartbeat
Worker
Worker
...
Worker
Cluster
Request New Resources & Reclaim Failure Resources
Data Processor
Model Trainer
Checkpoint Manager
HMP Training (§3)
Fine-grained Checkpoint (§4.1)
Efficient Reshard Loading (§4.2)
Efficient Saving (§5.3)


DP rank 0
LMBN = 18
DEVICES = [0, 8)
LAYERS = [0, 32)
DP rank 1
LMBN = 18
DEVICES = [8, 16)
LAYERS = [0, 32)
DP rank 2
LMBN = 15
DEVICES = [16)
LAYERS = [0, 4)
DEVICES = [18, 20)
LAYERS = [4, 13)
DEVICES = [20, 24)
LAYERS = [13, 32)
DP rank 3
LMBN = 13
DEVICES = [24, 26)
LAYERS = [0, 11)
DEVICES = [26, 30)
LAYERS = [11, 32)
0 1 2 3 4 5 6 7
8 9 10 11 12 13 14 15
16 17 18 19 20 21 22 23
24 25 26 27 28 29 30 31
Healthy GPU
Faulty GPU
TP Group
PP Comm